

Aplicações de Inteligência Artificial (41019, <https://www.ua.pt/en/uc/15297>)

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Project Development

Each group of two (or three) students is expected to work on one project topic. A group may

- (1) propose its own topic, subject to discussion with the instructors,
- (2) choose from the list of project proposals or topics based on ongoing research.

Every project should define a clear research question, use suitable data and evaluate its results against relevant methods or published findings.

Project Proposals

1. Skin Cancer Detection and Classification

International Skin Imaging Collaboration (ISIC)

Data: <https://challenge.isic-archive.com/data>

Challenges: <https://www.isic-archive.com/#!/topWithHeader/tightContentTop/challenges>

2. Face Detection from images

Github: <https://github.com/zlmo/Face-Detection>

<http://shuoyang1213.me/WIDERFACE/>

3. Face Recognition with Olivetti dataset

<https://www.kaggle.com/serkanpeldek/face-recognition-on-olivetti-dataset>

4. Machine Learning for cybersecurity

<https://github.com/PacktPublishing/Hands-on-Machine-Learning-for-Cyber-Security>

5. CNN for CIFAR-10 dataset

<https://www.cs.toronto.edu/~kriz/cifar.html>

6. Intel Image Scene Classification of Landscapes

Image Scene Classification of Multiclass (Buildings, Forest, Glacier, Mountain, Sea Street)

<https://www.kaggle.com/datasets/puneet6060/intel-image-classification>

7. AI4I 2020 Predictive Maintenance Data

Synthetic dataset that reflects real predictive maintenance encountered in industry.

<https://archive.ics.uci.edu/ml/datasets/AI4I+2020+Predictive+Maintenance+Dataset>

8. Garbage Classification and Image Processing

Image Classification (cardboard, glass, metal, paper, plastic, trash)

<https://www.kaggle.com/datasets/asdasdasdasdas/garbage-classification>

Topics based on ongoing research

9. Multi-step-ahead forecast error correction

Forecast errors often contain bias and temporal dependence, making multi-step-ahead predictions less reliable as the horizon increases. Recent literature proposed a hybrid approach combining LSTM neural networks with statistical bias correction to enhance 24-hour wind speed forecasts in Portugal [1]. The recursive multi-step-ahead forecast strategy corrects Global Forecast System (GFS) outputs, significantly reducing forecasting errors. This topic aims to investigate alternative strategies to forecast errors more accurately. Specifically, direct, multi-output or seq-to-seq error prediction methods can be explored as alternatives to the recursive approach, evaluating performance, breakdown in processing speed and residual autocorrelation analysis.

Data availability: A dataset consisting of historical hourly wind speed observations paired with numerical weather predictions from the GFS for Portugal will be provided upon request.

10. Stability of Deep Learning Anomaly Detection in Multivariate Time Series

Deep learning offers several approaches to anomaly detection in multivariate time series [2]. Among these, OmniAnomaly is a stochastic model that detects anomalies using reconstruction probabilities [3]. This topic investigates how its randomness affects the stability of anomaly scores, detected events and performance across repeated runs on the same data. The goal is to examine variation arising during training and inference, as well as the effect of the detection threshold, to assess how consistently anomalies are identified.

Data availability: The TSB-AD-M benchmark provides labelled multivariate time series and an OmniAnomaly implementation (<https://github.com/thedatumorg/TSB-AD>, [4]).

11. Improving Context Length Selection in Finite Context Models

Finite context models use the previous k symbols to predict the next one, making the choice of k central to compression performance. Recent literature proposed and evaluated criteria for selecting k based on partial auto mutual information (PAMI), but found that context selection remains difficult, particularly for shorter sequences and larger contexts [5, 6]. This topic aims to develop an improved method for choosing k , then estimates the smoothing parameter α conditional on that choice. Its compression performance and computational cost are compared with a genetic algorithm that searches jointly over k and α , using the same sequences and evaluation budget.

Data availability: NCBI Datasets Genome, <https://www.ncbi.nlm.nih.gov/datasets/genome>.

References

- [1] Gomes, V., Martins, A., Carvalho, D., & Gouveia, S. (2026). Recursive multi-step-ahead LSTM correction of GFS wind speed forecasts in Portugal. *Pattern Analysis and Applications*, 29(3). <https://doi.org/10.1007/s10044-026-01759-z>
- [2] Alves, B., Pinho, A. J., & Gouveia, S. (2026). *Unified taxonomy for multivariate time series anomaly detection using deep learning*. *Artificial Intelligence Review*. <https://doi.org/10.1007/s10462-026-11660-0>
- [3] Su, Y., Zhao, Y., Niu, C., Liu, R., Sun, W., & Pei, D. (2019). *Robust Anomaly Detection for Multivariate Time Series through Stochastic Recurrent Neural Network*. KDD. <https://doi.org/10.1145/3292500.3330672>
- [4] Liu, Q., & Paparrizos, J. (2024). The elephant in the room: Towards a reliable time-series anomaly detection benchmark. In *Advances in Neural Information Processing Systems* (Vol. 37, pp. 108231–108261). NeurIPS 2024. <https://doi.org/10.52202/079017-3437>
- [5] Contente, J., Martins, A., Pinho, A. J., & Gouveia, S. (2026). *A two-step sequential approach for hyperparameter selection in finite context models*. <https://doi.org/10.48550/arXiv.2603.19736>
- [6] Contente, J., Martins, A., Pinho, A. J., & Gouveia, S. (2026). *Comparing PAMI-Based Criteria for Context Length Selection in Finite-Context Models*. Submitted.